

Roll No.....

Odd End Term Examination, December-2025
Btech III Semester
Electrical and Electronics Measuring Instruments- TEE-303

Time: 3.00 Hrs

Maximum Marks: 100

Note:

- (i) Answer all the questions by choosing any two of the sub questions.
(ii) Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain working of RST flip flop with its circuit diagram, working and output waveform.	CO-01 & CO-02
(b)	Describe the working of digital frequency meter with the help of its block diagram.	
(c)	A balanced delta connected load of $(8+j6)\Omega$ per phase is connected to a balanced 3-phase 400 V supply. Find the line current, power factor, power and total volt amperes.	
Q2	(20 marks)	
(a)	Explain the working of Schmitt Trigger.	CO-02 & CO-03
(b)	Explain Op-Amps and their transfer characteristics.	
(c)	In the case of a Schering Bridge, arm AC has $R=4.7k\Omega$. Arm CD has unknown elements. Arm BD has $C=0.1\mu F$ Arm AB= $4.7K\Omega$ is shunt with IMF. Determine Values of components is the arm CD.	
Q3	(20 marks)	
(a)	Explain the construction and working of a cathode ray tube with a neat diagram.	CO-01 & CO-03
(b)	Explain two wattmeter method for star interconnection with appropriate phasor diagram.	
(c)	Explain Inverting mode of Op-Amp.	
Q4	(20 marks)	

(a)	A current transformer has a single turn primary and a 200 turns secondary winding. The secondary winding supplies a current of 5A to a non-inductive burden of 1Ω resistance. The requisite flux is set up in the core by an mmf of 80A. The frequency is 50 Hz and the net cross section of the core is 1000 mm^2 . Calculate the ratio and phase angle of the transformer. Also find the flux density in the core. Neglect the effects of magnetic leakage, iron losses and I^2R losses.	CO-04
(b)	Explain Potential Transformer in detail with the help of phasor diagram.	
(c)	What are Lissajous pattern? Explain their use with respect to CRO.	
Q5	(20 marks)	
(a)	Explain Maxwell's inductance bridge with its circuit diagram, phasor diagram and also derive the expression of inductance measured with the help of this bridge.	
(b)	A basic d'arsonval meter movement with an internal resistance $R_m=100\Omega$, and a full scale a current of $I_m=1\text{mA}$, is to be converted into a multi range d.c.voltmeter with ranges of 0-10V, 0-50V, 0-250V and 0-500V. Find the values of various resistances using the potential divider arrangement.	CO-04&05
(c)	What is the difference between analog and digital instruments?	